

Practical 15 - Multinomial and Ordinal Logistic Regression

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `multinomial.csv`

Introduction

In this practical we consider data from the 2003-2004 US National Health and Nutrition Examinations Survey (NHANES), fitting models that relate a categorised version of systolic blood pressure (SBP) to age and sex. We will categorise blood pressure according to SBP as follows.

Category	SBP
1. Normal	< 120
2. Prehypertension	$120 - 139$
3. High (stage 1)	$140 - 159$
4. High (stage 2)	≥ 160

Given the above classification system, it is of interest to model how the probability of falling into the four categories depends on age and sex.

Data

The dataset is called `multinomial.csv` and contains three variables as below.

Variable	Description
sex	“Female”, “Male”
ageyrs	Age in (whole number of) years
sbp	Systolic blood pressure (mmHg)

Aims

- To understand how to interpret coefficients from multinomial and ordinal logistic regression models.
- To understand how to calculate estimated probabilities from multinomial and ordinal logistic regression models.
- To understand how to compare the fits of multinomial and ordinal logistic regression models.

Analysis

Question 1

Import the data into Stata, and use **encode** so that the **sex** variable can be used as a covariate later in regression models.

Create a new variable for SBP, categorised as above. One way to do this in Stata is as follows.

```
egen sbpcat = cut(sbp), at(0 120 140 160 300) icodes
```

```
replace sbpcat = sbpcat + 1
```

Tabulate **sbpcat** to see its distribution.

Also, to aid interpretation of the intercept estimates in our models, centre the age variable at 50 years.

```
gen cen_age = ageyrs - 50
```

SOLUTIONS

```
. import delimited multinomial.csv
. egen sbpcat=cut(sbp), at(0 120 140 160 300) icodes
. replace sbpcat = sbpcat + 1
```

Here the `icodes` option tells `egen` to code the new variable as 0, 1, 2 etc., rather than its default, which is to use the left-hand interval values of the categories. The `replace` command codes the new variable as 1, 2, 3, 4 rather than 0, 1, 2, 3 so as to match the lecture notation.

```
. tab sbpcat
```

	SBP	Freq.	Percent	Cum.
Normal		1,220	47.88	47.88
Prehypertension		896	35.16	83.05
High (stage 1)		287	11.26	94.31
High (stage 2)		145	5.69	100.00
Total		2,548	100.00	

Most participants are in the 'Normal' category, with the numbers in each successive category becoming progressively smaller.

Question 2

Fit a multinomial logistic model to the categorised SBP variable with sex and centred age as covariates.

Carefully describe the interpretation of each of the estimated parameters.

SOLUTIONS

```
. mlogit sbpcat i.sex cen_age
```

Multinomial logistic regression	Number of obs = 2,548
	LR chi2(6) = 732.11
	Prob > chi2 = 0.0000
Log likelihood = -2511.1605	Pseudo R2 = 0.1272

sbpcat	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Normal	(base outcome)					
Prehypertension						
sex						
Male	.6070803	.0945075	6.42	0.000	.421849	.7923116

cen_age		.0422103	.0029177	14.47	0.000	.0364917	.047929
_cons		-.3877347	.0734775	-5.28	0.000	-.5317479	-.2437215
-----+-----							
High__stage_1_							
sex							
Male		.3098523	.1441437	2.15	0.032	.0273359	.5923687
cen_age		.0786693	.0044112	17.83	0.000	.0700235	.0873151
_cons		-1.540525	.1117048	-13.79	0.000	-1.759463	-1.321588
-----+-----							
High__stage_2_							
sex							
Male		-.3101371	.1938604	-1.60	0.110	-.6900964	.0698223
cen_age		.1067755	.0065561	16.29	0.000	.0939257	.1196252
_cons		-2.313477	.1582206	-14.62	0.000	-2.623584	-2.00337
-----+-----							

First note that `mlogit` by default takes the most common outcome level as the base (reference) outcome level. This can be modified if desired using the `baseoutcome()` option.

-.3877347 is the estimated log odds of being in the ‘Prehypertension’ category in females aged 50, ignoring females in the ‘High (stage 1)’ and ‘High (stage 2)’ categories.

0.6070803 is the estimated log odds ratio for having systolic blood pressure in the ‘Prehypertension’ category comparing males with females, conditional on age, and ignoring people in the ‘High (stage 1)’ and ‘High (stage 2)’ categories. If you specify the `rrr` option to exponentiate the coefficients, Stata instead refers to this as a relative risk ratio, since it is the ratio of two relative risks: the numerator relative risk is the risk of being in the ‘Prehypertension’ category divided by the risk of being in the ‘Normal’ category in a male whilst the denominator relative risk is the risk of being in the ‘Prehypertension’ category divided by the risk of being in the ‘Normal’ category in a female of the same age.

.0422103 is the estimated log odds ratio for having systolic blood pressure in the ‘Prehypertension’ category for a one-year increase in age, conditional on sex, and ignoring people in the ‘High (stage 1)’ and ‘High (stage 2)’ categories.

The interpretation of the three ‘High (stage 1)’ parameter estimate are analogous to those given above. Here, however, all of the comparisons are between the ‘High (stage 1)’ and ‘Normal’ categories, ignoring people in the ‘Prehypertension’ and ‘High (stage 2)’ categories.

The interpretation of the three ‘High (stage 2)’ parameter estimate are also analogous to those given above. Here, however, all of the comparisons are between the ‘High (stage 2)’ and ‘Normal’ categories, ignoring people in the ‘Prehypertension’ and ‘High (stage 1)’ categories.

Question 3

From your fitted model, calculate (by hand) the estimated probability of having SBP in the second highest category (`spbcat = 3`) for a female aged 70 years.

SOLUTIONS

The probability that a female aged 70 (i.e. with centred age = 20) has systolic blood pressure in the *High (stage 1)* category is (using the formulae in the lecture notes)

$$\frac{\exp(-1.540525 + 20 \times 0.078669)}{1 + \exp(-.3877347 + 20 \times 0.042210) + \exp(-1.540525 + 20 \times 0.078669) + \exp(-2.313477 + 20 \times 0.106776)} =$$

Note that the calculation requires the use of **all three linear predictors**, not just the linear predictor reported for the *High (stage 1)* category.

Question 4

Use the `predict` command to generate predicted probabilities of individuals being in each of the four SBP categories, and plot the probabilities against age, separately in males and females. Hint: in order to see the lines sort the data by age before plotting the probabilities.

Check that the predicted probability of a 70 year old female having SBP in the second highest category (`spbcat = 3`) matches that calculated above.

SOLUTIONS

```
. predict pr1 pr2 pr3 pr4

. sort ageyrs

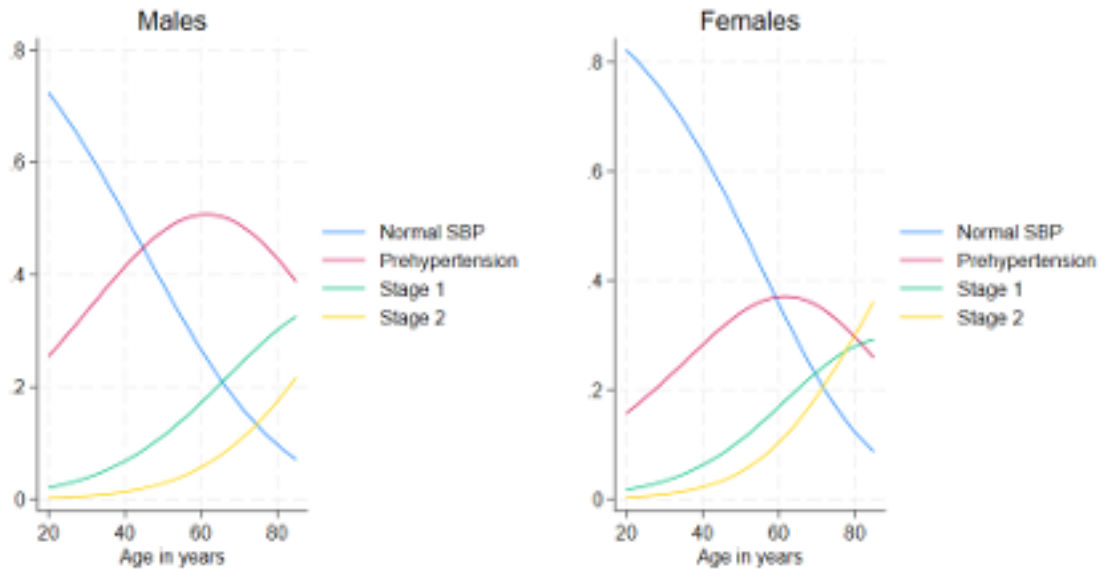
. line pr1 pr2 pr3 pr4 ageyrs if sex==1, title("Males") ///
legend(order( 1 "Normal SBP" 2 "Prehypertension" 3 "Stage 1" ///
4 "Stage 2") cols(1) colfirst) sort saving(g1, replace)

. line pr1 pr2 pr3 pr4 ageyrs if sex==2, title("Females") ///
legend(order( 1 "Normal SBP" 2 "Prehypertension" 3 "Stage 1" ///
4 "Stage 2") cols(1) colfirst) sort saving(g2, replace)

. graph combine "g1" "g2", title("Multinomial logistic regression /// model") subtitle("Pred.
```

Multinomial logistic regression model

Predicted probabilities by gender and age



Notice that the way in which the fitted probabilities of being in each category change with increasing age is not always monotonic. The multinomial logistic regression model does not preclude non-monotonic dependencies on covariates.

```
.      summ pr3 if cen_age==20 & sex==1
```

Variable	Obs	Mean	Std. dev.	Min	Max
-----+-----					
pr3	9	.2322852	0	.2322852	.2322852

The predicted probability that a female aged 70 (i.e., with centred age = 20) has systolic blood pressure that is in the 'High (stage 1)' category matches the calculation in question 3.

Question 5

Now fit an ordinal logistic regression model to the categorised SBP variable with sex and centred age as covariates.

Carefully describe the interpretation of each of the estimated parameters.

SOLUTIONS

```
. ologit sbpcat i.sex cen_age
```

Ordered logistic regression

Number of obs = 2,548

LR chi2(2) = 675.85

Prob > chi2 = 0.0000

Log likelihood = -2539.288

Pseudo R2 = 0.1174

sbpcat	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
+						
sex						
Male	.2795258	.0795941	3.51	0.000	.1235243	.4355274
cen_age	.0574355	.002395	23.98	0.000	.0527415	.0621296
+						
/cut1	-.1949374	.0634314			-.3192607	-.0706141
/cut2	1.87444	.0747071			1.728017	2.020863
/cut3	3.24833	.1019183			3.048573	3.448086

The estimate .0574355 is the estimated (common) log odds ratio for a one-year increase in age conditional on sex. Under the ordinal logistic model this odds ratio is simultaneously

- i) the log odds ratio for **sbpcat** equalling 2 or more rather than 1,
- ii) the log odds ratio for **sbpcat** equalling 3 or more rather than 2 or less, and
- iii) the log odds ratio for **sbpcat** equalling 4 rather than 3 or less.

All three of these log odds ratios are assumed to be equal (the proportional odds assumption).

From the course manual, we know that the estimated cut-points are the estimates of minus the intercept parameters. Thus, +.1949374 is the estimated log odds of **sbpcat** equalling 2 or more for a female aged 50 (the age of someone for whom the centred age variable is zero). And 1.87444 is the estimated log odds of **sbpcat** equalling 3 or more for a female aged 50, whilst -3.24833 is the estimated log odds of **sbpcat** equalling 4 for a female aged 50.

Question 6

From your fitted ordinal logistic regression model, calculate (by hand) the estimated probability of having SBP in the second highest category (**sbpcat** = 3) for a female aged 70 years.

SOLUTIONS

Using the parameter estimates from the ordinal logistic regression model, we can estimate the probability that a female aged 70 (i.e. with centred age = 20) has systolic blood pressure in the *High (stage 1)* category, **or a higher category**, as follows:

$$\frac{\exp(-1.87444 + 20 \times 0.0574355)}{1 + \exp(-1.87444 + 20 \times 0.0574355)} = 0.326$$

We can also estimate the probability that a female aged 70 has systolic blood pressure in the *High (stage 2)* category, or higher, as follows:

$$\frac{\exp(-3.24833 + 20 \times 0.0574355)}{1 + \exp(-3.24833 + 20 \times 0.0574355)} = 0.109$$

Therefore, we can estimate the probability that a female aged 70 has systolic blood pressure in the *High (stage 1)* category by subtraction: $0.326 - 0.109 = 0.217$.

The predicted probability here (0.217) is not very different from that calculated in question 3 for the multinomial logistic regression model (0.232).

Question 7

Use the `predict` command to generate predicted probabilities of individuals being in each of the four SBP categories for the ordinal logistic regression model, and plot the probabilities against age, separately in males and females.

Again check that the predicted probability of a 70 year old female having SBP in the second highest category (`spbcat = 3`) matches that calculated above.

Are the predictions from the ordinal and multinomial logistic regression models markedly different?

SOLUTIONS

```
. predict pr5 pr6 pr7 pr8
```

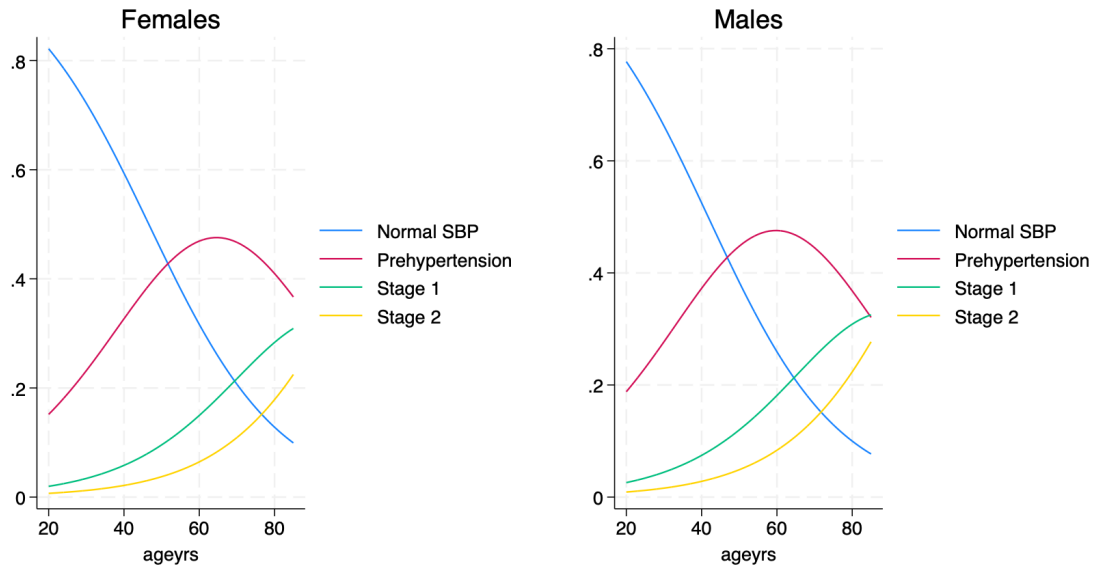
```
.          summ pr7 if cen_age==20 & sex==1
```

Variable	Obs	Mean	Std. dev.	Min	Max
pr7	9	.2169988	0	.2169988	.2169988

The predicted probability that a female aged 70 (i.e., with centred age = 20) has systolic blood pressure that is in the ‘High (stage 1)’ category matches the calculation in question 6.

Ordinal logistic regression model

Predicted probabilities by sex and age



As with the multinomial logistic regression model, the way in which the fitted probabilities of being in each category change with increasing age is not always monotonic. With the ordinal logistic regression model the fitted probabilities of being in the highest and lowest categories are always monotonic functions of the linear predictor, but the fitted probabilities of being in the intermediate categories need not be.

Comparing this graph with the analogous plot for the ordinal logistic regression model suggests some differences, these being most obvious for older females.

Question 8

To further assess whether the proportional odds assumption is reasonable here compare the log-likelihoods and AIC values from the two models.

SOLUTIONS

```
. qui mlogit sbpcat i.gender cen_age
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	2,548	-2877.215	-2511.161	9	5040.321	5092.909

Note: BIC uses N = number of observations. See [R] IC note.

```
. qui ologit sbpcat i.gender cen_age
```

```
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	2,548	-2877.215	-2539.288	5	5088.576	5117.791

Note: BIC uses N = number of observations. See [R] IC note.

The AIC is markedly lower (better) for the multinomial logistic regression model.

Question 9

As a final check of the ordinal logistic regression model fit, compare the coefficients to those from a series of logistic regression models with different cut-offs. First create three new variables, to dichotomise the SBP variable in three different ways:

1. Normal vs All other categories
2. Normal or Prehypertension vs Hypertension (Stage 1 or 2)
3. All other categories vs Stage 2 Hypertension

Fit separate standard logistic regression models for each of these new variables as the outcome, and sex and centred age as covariates. Compare the coefficients from each model to the equivalent from the ordinal logistic regression model.

Considering all of the above, which model do you prefer for these data?

SOLUTIONS

Logistic regression

Number of obs = 2,548

LR chi2(2) = 526.21

Prob > chi2 = 0.0000

Log likelihood = -1500.7444

Pseudo R2 = 0.1492

sbpcat1	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
-----+-----						
sex						
Male	.4924389	.0887338	5.55	0.000	.3185238	.666354
cen_age	.0542514	.0027301	19.87	0.000	.0489006	.0596022
_cons	.0460239	.0662779	0.69	0.487	-.0838784	.1759261

. logit sbpcat2 i.sex cen_age, nolog

Logistic regression

Number of obs = 2,548

LR chi2(2) = 424.80

Prob > chi2 = 0.0000

Log likelihood = -947.35477

Pseudo R2 = 0.1831

sbpcat2	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
-----+-----						
sex						
Male	-.2270577	.1173106	-1.94	0.053	-.4569823	.0028669
cen_age	.0639055	.0034962	18.28	0.000	.057053	.0707579
_cons	-1.670222	.089143	-18.74	0.000	-1.844939	-1.495505

. logit sbpcat3 i.sex cen_age, nolog

Logistic regression

Number of obs = 2,548

LR chi2(2) = 212.78

Prob > chi2 = 0.0000

Log likelihood = -450.01946

Pseudo R2 = 0.1912

-----+-----						
sbpcat3	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
-----+-----						
sex						
Male	-.6652546	.1830939	-3.63	0.000	-1.024112	-.3063971
cen_age	.0720945	.0058268	12.37	0.000	.0606743	.0835147

_cons		-2.983977	.1485475	-20.09	0.000	-3.275125	-2.69283
-------	--	-----------	----------	--------	-------	-----------	----------

The three estimated log odds ratios for a one unit increase in age are 0.054 (95% CI 0.049 to 0.060), 0.064 (0.057 to 0.071) and 0.072 (0.061 to 0.084) - some variation, but not not hugely different from one another or from the estimate obtained assuming a common log odds ratio (0.057 (0.053 to 0.062)). However, the three estimated log odds ratios for the comparison between males and females vary markedly: 0.49 (0.67 to 0.32), -0.23 (0.00 to -0.46) and -0.66 (-0.31 to -1.02). There is evidence that conditional on age, males are less likely to be in the highest category of blood pressure vs. the lower levels, and also more likely to be in the lowest category of blood pressure vs. the higher levels.

The comparisons we have made in questions 8 and 9 suggest that the multinomial logistic regression model better describes the data here. The predicted probabilities from the two models are not hugely different different, but nonetheless we would probably prefer those from the multinomial logistic regression model.